

Tutorial - 1 - Answers

1. (a) Static.

(b) Dynamic.

2. (a) non-causal.

(b) causal (for $a > 0$); non-causal (for $a < 0$).

(c) causal.

3. (a) Time varying.

(b) Time-invariant.

4. (a) nonlinear.

(b) nonlinear.

(c) linear.

8. $Y(s) = \frac{P(s)C(s)}{1 + P(s)C(s)F(s)}R(s); U(s) = \frac{C(s)}{1 + P(s)C(s)F(s)}R(s); E(s) = \frac{1}{1 + P(s)C(s)F(s)}R(s).$

9. $\frac{Y(s)}{U(s)} = \frac{G_1(s)G_2(s)G_4(s)G_5(s)}{1 + G_1(s)G_2(s)G_3(s) + G_4(s)G_5(s) + G_2(s)G_4(s) + G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)}.$